EXPERIMENT NO: 09

TITLE: Communicate between Arduino and Raspberry PI using any wireless medium like ZigBee

AIM/ OBJECTIVES: To establish wireless communication between Arduino and Raspberry Pi using ZigBee modules for data transmission and reception.

REQUIREMENT:

SOFTWARE: XCTU software (for configuring XBee modules), Arduino IDE,Python (on Raspberry Pi)

HARDWARE/EQUIPMENT: Arduino Uno (or compatible Arduino board),Raspberry Pi (any model with GPIO access),2 × XBee Series 2 modules (ZigBee communication),2 × XBee Shields or USB adapters (for Arduino and Raspberry Pi),Jumper wires,USB cables (for programming),Arduino and Raspberry Pi),External power source (if required)

THEORY:

ZigBee is a low-power, low-data-rate wireless communication protocol based on IEEE 802.15.4, widely used in IoT, sensor networks, and home automation. XBee modules implement ZigBee, supporting point-to-point, star, and mesh networking.
In this experiment:

- **Arduino** acts as the transmitter (End Device or Router).
- **Raspberry Pi** acts as the receiver (Coordinator).
- Data is exchanged wirelessly via configured XBee modules.

1. Configuring XBee Modules using XCTU:

1. Install **XCTU software** from Digi's website.
2. Connect XBee modules via USB adapters or shields.
3. Launch XCTU, detect the modules.
4. Configure:
    - One XBee as **Coordinator** (connect to Raspberry Pi).
    - Other XBee as **End Device/Router** (connect to Arduino).
    - Assign the **same PAN ID** for both modules
    - Set each module's **destination address** to the other module.
5. Test communication in XCTU by sending messages between modules.

2. Hardware Setup:

Arduino with XBee:

- Connect XBee Shield to Arduino.
- Place XBee module on the shield.
- Connect Arduino to PC via USB.
- XBee communicates via **pins D0 (RX) and D1 (TX).**

Raspberry Pi with XBee:

- Connect XBee via USB adapter or GPIO (TX, RX, 3.3V, GND).
    - **TX of XBee → RX of Raspberry Pi**
    - **RX of XBee → TX of Raspberry Pi**
    - Connect 3.3V and GND properly.

Access Raspberry Pi via SSH or monitor/keyboard setup

3. Arduino Code:

```
#include <SoftwareSerial.h>

SoftwareSerial XBee(2, 3); // RX, TX pins for XBee void

setup() {
  Serial.begin(9600);            // Serial communication with PC
  XBee.begin(9600);              // Serial communication with XBee
}

void loop() {
  XBee.println("Hello from Arduino"); // Send data every second delay(1000);
}
```

Connections:

- Arduino D2 → XBee RX
- Arduino D3 → XBee TX

4. Raspberry Pi Setup:

Step 1: Install Python Serial Library

```
sudo apt update
sudo apt install python3-pip
pip3 install pyserial
```

Step 2: Python Code to Receive Data:

```
import serial
# Replace '/dev/ttyUSB0' with your XBee port xbee =
serial.Serial('/dev/ttyUSB0', 9600)
while True:
    if xbee.in_waiting > 0:
        incoming_data = xbee.readline().decode('utf-8').strip() print(f"Received from
        Arduino: {incoming_data}")
```

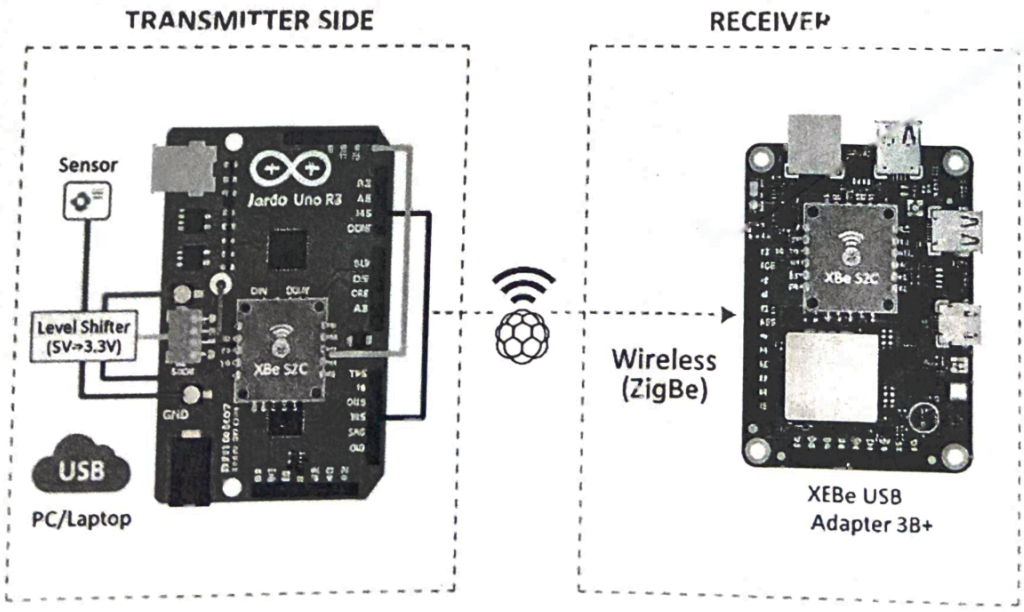
- Run the script:

```
python3
xbee_receive.py
```

5. Testing Communication:

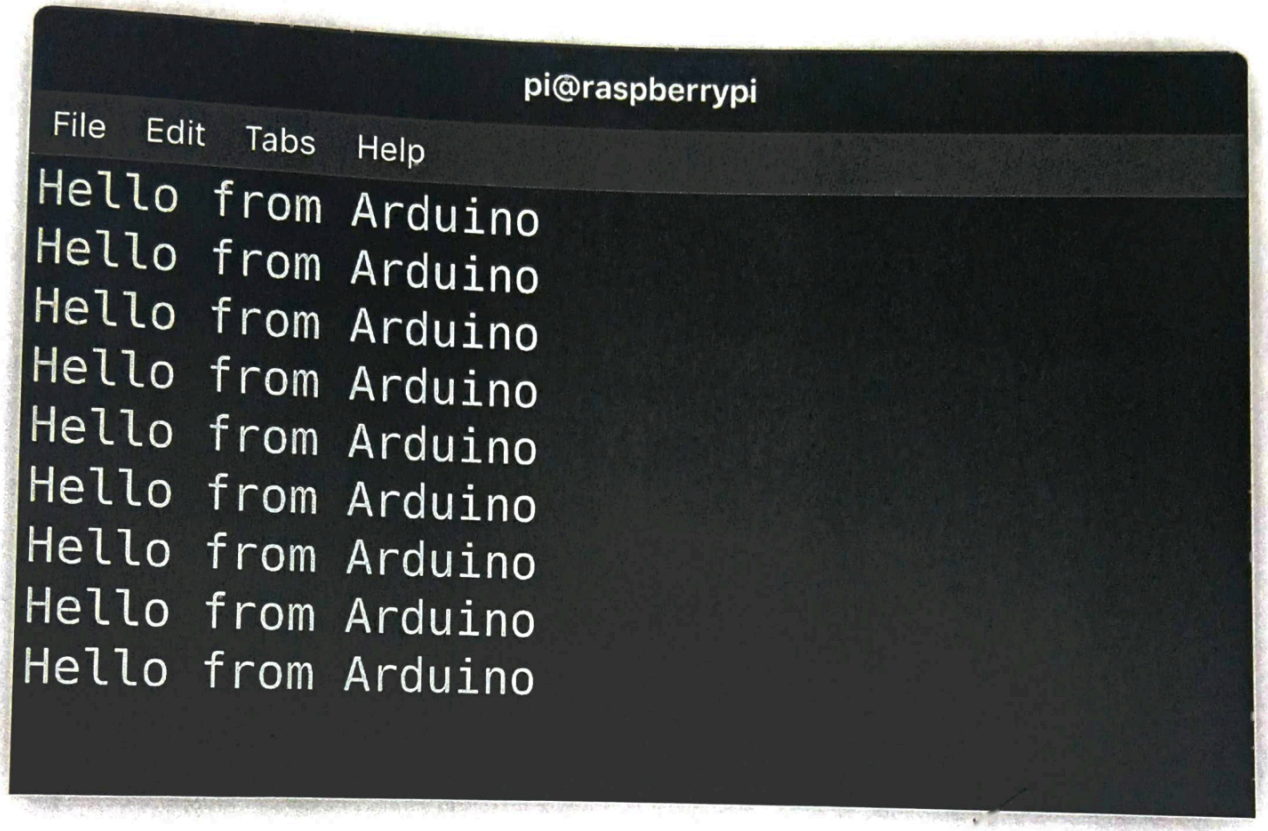
1. Power Arduino and Raspberry Pi with XBee modules.
2. Run the Python script on Raspberry Pi.
3. Arduino sends **"Hello from Arduino"** every second.
4. Raspberry Pi terminal displays the received message.

CIRCUIT/BLOCK DIAGRAM:



Ex-9_Xbee | Arduino 1.8.19
File Edit Sketch Tools Help
Ex-9_Xbee
#include <SoftwareSerial.h>
SoftwareSerial XBee(2, 3); // RX, TX pins for XBee
void setup() {
Serial.begin(9600); // Serial communication with PC
XBee.begin(9600); // Serial communication with XBee
}
void loop() {
XBee.println("Hello from Arduino"); // Send data every second
delay(1000);
}

Done compiling.
Sketch uses 3254 bytes (10%) of program storage space. Maximum is 32256 bytes.
Global variables use 323 bytes (15%) of dynamic memory, leaving 1725 bytes for local variables. Maximum is 2048 bytes.
Type here to search


pi@raspberrypi
File Edit Tabs Help
Hello from Arduino
Hello from Arduino
Hello from Arduino
Hello from Arduino
Hello from Arduino
Hello from Arduino
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Hello from Arduino
Hello from Arduino